Foundations of Artificial Intelligence - 2026.1

Atomic Decisions under Uncertainty

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Decision Making Situations

Decision Situation Matrix

Action Effects

Horizon	**Certainty**	**Risk**	**Uncertainty**
Atomic	✓	✓	NOW
Sequential	✓	✓	

Bandits

Motivation

Consider this problem: Every day you can choose between cafeteria 1 and 2, each of which offers only one menu. The menus are different every day and meet your taste with a score between 1 and 5. Once you went to one cafeteria, you cannot change your mind.

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... you should pick the cafeteria with the highest expected utility (average rating).

The problem is that you do not know in advance which meal is served by which cafeteria that day, and since you are new to the place, you also don't know with which probability they are served.

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On the first day, we cannot make any rational justified choice.

But over time we will learn in which cafeteria we receive better meals, on average.

Effectively, what we do is to convert a decision of uncertainty into one under risk.

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Problem Definition

Consider an infinite *episodic* game in which we can choose, in each round $t = 1, 2, ..,$ an action $A_t \in \{1, .., K\}$, each of which associated with an unknown reward distribution $\mathbb{P}_i$.

Choosing action A_t at time t samples a reward $R_t \sim \mathbb{P}_{A_t}$.

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Challenge: Since we do not know $\mathbb{P}_a$, we do not know $q(a)$.

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Problem Definition - Offline Scenario

If we knew $\mathbb{P}_a$ for all arms, we could compute or at least approximate $q(a)$.

Then picking the optimal action would be easy.

We could try to learn $\mathbb{P}_a$, but since this is only used for $q(a)$, we can learn $q(a)$ directly.

So we could give some finite time horizon T to learn the true values $q(a)$ and then always use the arm we think is best.

Bandits

Estimating Average Reward

How to estimate $q(a)$ based on choice A_t and observations R_t up to time t?

An intuitive way is to take the mean of rewards observed for arm a up to time t:

$$\boxed{\hat{q}_t(a) := \frac{\sum_{s=1}^{t-1} R_s \cdot 1_{A_s=a}}{\sum_{s=1}^{t-1} 1_{A_s=a}}}$$

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3. it quickly concentrates around $q(a)$: $\mathbb{V}[\hat{q}_t(a)] = \mathbb{V}[\mathbb{P}_a]/t$

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Estimating Average Reward

A nice propety of the empirical mean is that it can be updated incrementally:

$$\boxed{\hat{q}_n(a) := \hat{q}_{n-1}(a) + \frac{1}{n}[R_n - \hat{q}_{n-1}(a)]}$$

where we simplified notation a bit to say that n is the time n-th time step in which action a was chosen.

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This way, the time to compute $\hat{q}_n(a)$ is constant and does not grow with n.

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Exploration vs. Exploitation

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As soon as $\hat{q}_t(a) \approx q(a)$, we identify a^* and we can exploit, i.e. only draw a^*.

But how do we know whether $\hat{q}_t(a) \approx q(a)$?

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Balancing Exploitation and Exploration

It is *impossible* to *ever* be sure that one has identified the best arm.

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Intuitively: Pull $\arg\max_a \hat{q}_t(a)$ but maintain some mechanism for exploration.

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Problem Definition

Any strategy we may use creates a (usually stochastic) sequence $(A_t)_{t \in \mathbb{N}}$.

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The latter two can only be computed if we (the experimenters) actually know the $\mathbb{P}_a$.

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The Greedy Strategy

The simplest strategy is to try each arm once and then always pull $\arg\max_a \hat{q}_t(a)$.

The Greedy Strategy is exploitation-only, virtually there is no exploration beyond initialization.

This is ideal if $a^* = \arg\max_a \hat{q}_t(a)$ but can be pretty bad otherwise.

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The ε-greedy Strategy

A simple modification to consider both aspects is to define some exploration probability ε_t as a function of time t (but independent of the history).

At time t, one pulls the arm

$$\arg\max_a \ \hat{q}_t(a)$$

with probability $1 - \varepsilon_t$ and with probability ε_t a random arm.

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Note that $\varepsilon_t = 0$ is the greedy strategy.

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The UCB1 Policy

A simple but effective bandit strategy is UCB1. Here, the confidence bound is created using a factor that balances the number of pulls of an arm in terms of the overall time elapsed so far:

$$\boxed{A_t^{UCB} := \arg\max_{a \in \{1 \dots K\}} \left\{ \hat{q}_t(a) + c\sqrt{\frac{\ln(t)}{N_t(a)}} \right\},}$$

where

- c is an exploration hyperparameter, often set to $\sqrt{2}$ if we know that $R_t \in [0, 1]$,
- $N_t(a)$ counts how often a has been tried at time t.

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The UCB1-Policy

The second term $c\sqrt{\frac{\ln t}{N_t(a)}}$ is often called exploration term.

Properties of the exploration term are:

- proportional to the time t and inversely proportional to the sample size $N_t(a)$.
- Expands slightly over time and contracts strongly for new observations.
  $\Rightarrow$ For big values we have little confidence in the estimate and inverse.
- We can hence understand this as an exploration term since it controls under which conditions we should pull an arm that appears sub-optimal.

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Asymptotic Optimality

Since every (non-oracle) policy must explore, it is not possible that the contributions to the regret drop to 0 permanently:

Even if we already identified the best arm, we still must explore every once in a while, and this will augment the regret.

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Most strategies (including UCB1) satisfy this condition, and *every* strategy that keeps exploring eventually identifies the best option. However, the question is how they behave in the *short* run.

The Influence of Initial Values

How do we initialize $\hat{q}(a)$?

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What are the effect of all of these on exploration (with UCB1)?

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Summary

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  Therefore we must interleave and trade off exploitation (use of knowledge) with exploration (complement knowledge)
- all strategies that do this must have a logarithmic regret, but some are better than others, i.e., UCB1 better than greedy-ε